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TREND-BASED MOCK PAPER

GATE Life Sciences

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Life Sciences
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section B: Life Sciences

Q1. [1-MCQ] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q2. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q3. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q4. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q5. [1-NAT] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q6. [1-NAT] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q7. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
 - (B) Positive definite
 - (C) Diagonally dominant
 - (D) Non-singular
-

Q8. [1-NAT] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
 - (B) $y = C/x$
 - (C) $y = C \cdot x^2$
 - (D) $x^2 + y^2 = C$
-

Q10. [1-MCQ] [1 mark]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
 - (B) $1/(s-2)^2$
 - (C) $s/(s+2)^2$
 - (D) $2/(s+2)^2$
-

Q11. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q12. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q13. [2-NAT] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q14. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-NAT] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q17. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q18. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q19. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1,0]^T$ equals:

- (A) $[1,0]^T$
- (B) $[2,1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q20. [2-MSQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q21. [2-MSQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q22. [2-NAT] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-NAT] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-NAT] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q26. [2-NAT] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

Fourier series of $f(x) = x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
 - (B) Cosine terms
 - (C) Both
 - (D) Neither
-

Q28. [1-NAT] [1 mark]

Strongest field ligand (spectrochemical series):

(Numerical Answer Type — Enter the numerical value)

Q29. [1-MCQ] [1 mark]

$[\text{Fe}(\text{CN})_6]^{4-}$ (low spin, d^6) is:

- (A) Diamagnetic
 - (B) Paramagnetic
 - (C) Ferromagnetic
 - (D) Antiferromagnetic
-

Q30. [1-MCQ] [1 mark]

Octahedral crystal field splits d-orbitals into:

- (A) t_{2g} and e_g
 - (B) e_g and t_{2g} only
 - (C) s and p
 - (D) sigma and pi
-

Q31. [1-MCQ] [1 mark]

Werner theory explains:

- (A) Bonding in complexes
- (B) CN and geometry
- (C) Magnetic properties

(D) Color only

Q32. [2-MCQ] [2 marks]

[Co(NH₃)₆]³⁺ coordination number:

- (A) 3
- (B) 4
- (C) 6
- (D) 8

Q33. [2-NAT] [2 marks]

Werner theory explains:

(Numerical Answer Type — Enter the numerical value)

Q34. [2-MCQ] [2 marks]

Strongest field ligand (spectrochemical series):

- (A) I⁻
- (B) Br⁻
- (C) CN⁻
- (D) F⁻

Q35. [2-MCQ] [2 marks]

[Co(NH₃)₆]³⁺ coordination number:

- (A) 3
- (B) 4
- (C) 6
- (D) 8

Q36. [2-MCQ] [2 marks]

Strongest field ligand (spectrochemical series):

- (A) I⁻
- (B) Br⁻
- (C) CN⁻
- (D) F⁻

Q37. [2-NAT] [2 marks]

Octahedral crystal field splits d-orbitals into:

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MCQ] [2 marks]

Werner theory explains:

- (A) Bonding in complexes
- (B) CN and geometry
- (C) Magnetic properties
- (D) Color only

Q39. [1-MCQ] [1 mark]

Strongest acid:

- (A) HCl
- (B) HNO₃
- (C) H₂SO₄
- (D) HClO₄

Q40. [1-NAT] [1 mark]

pH of 10^{-8} M HCl approx:

(Numerical Answer Type — Enter the numerical value)

Q41. [1-NAT] [1 mark]

Galvanic cell converts:

(Numerical Answer Type — Enter the numerical value)

Q42. [1-NAT] [1 mark]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

(Numerical Answer Type — Enter the numerical value)

Q43. [1-MCQ] [1 mark]

First-order half-life with $k = 0.693 \text{ min}^{-1}$:

- (A) 1 min
 - (B) 0.693 min
 - (C) 1.44 min
 - (D) 0.5 min
-

Q44. [2-MCQ] [2 marks]

Strongest acid:

- (A) HCl
 - (B) HNO_3
 - (C) H_2SO_4
 - (D) $HClO_4$
-

Q45. [2-NAT] [2 marks]

pH of 10^{-8} M HCl approx:

(Numerical Answer Type — Enter the numerical value)

Q46. [2-MCQ] [2 marks]

Galvanic cell converts:

- (A) Electrical to chemical
 - (B) Chemical to electrical
 - (C) Heat to work
 - (D) Mechanical to electrical
-

Q47. [2-MCQ] [2 marks]

pH of 10^{-8} M HCl approx:

- (A) 6
 - (B) 7
 - (C) 8
 - (D) 6.98
-

Q48. [2-MSQ] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q49. [1-MCQ] [1 mark]

$\Delta G = -5 \text{ kJ/mol}$ at 300K. Equilibrium constant:

- (A) e^2
- (B) e^{-2}
- (C) 2
- (D) 0.5

Q50. [1-MCQ] [1 mark]

Rate doubles per 10K rise. Activation energy ($T=300\text{K}$):

- (A) $\sim 110 \text{ kJ/mol}$
- (B) $\sim 56 \text{ kJ/mol}$
- (C) $\sim 22 \text{ kJ/mol}$
- (D) $\sim 8 \text{ kJ/mol}$

Q51. [1-NAT] [1 mark]

pH of 0.01M HCl:

(Numerical Answer Type — Enter the numerical value)

Q52. [1-MCQ] [1 mark]

First-order: 50% in 10 min. 90% time:

- (A) 10 min
- (B) 33.2 min
- (C) 20 min
- (D) 5 min

Q53. [1-MCQ] [1 mark]

Van der Waals corrects for:

- (A) Only volume
- (B) Only pressure
- (C) Both volume and pressure
- (D) Temperature

Q54. [2-NAT] [2 marks]

pH of 0.01M HCl:

(Numerical Answer Type — Enter the numerical value)

Q55. [2-MCQ] [2 marks]

First-order: 50% in 10 min. 90% time:

- (A) 10 min
- (B) 33.2 min
- (C) 20 min
- (D) 5 min

Q56. [2-MCQ] [2 marks]

$\Delta G = -5 \text{ kJ/mol}$ at 300K. Equilibrium constant:

- (A) e^2
- (B) e^{-2}
- (C) 2
- (D) 0.5

Q57. [2-NAT] [2 marks]

First-order: 50% in 10 min. 90% time:

(Numerical Answer Type — Enter the numerical value)

Q58. [2-MCQ] [2 marks]

Van der Waals corrects for:

- (A) Only volume
 - (B) Only pressure
 - (C) Both volume and pressure
 - (D) Temperature
-

Q59. [1-NAT] [1 mark]

Hofmann elimination gives:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

Grignard reacts with:

- (A) Alcohol only
 - (B) CO₂, carbonyl compounds
 - (C) Acids only
 - (D) Alkanes
-

Q61. [1-MCQ] [1 mark]

Geometrical isomerism in:

- (A) 1-butene
 - (B) 2-butene
 - (C) 1-butyne
 - (D) 2-butyne
-

Q62. [1-NAT] [1 mark]

S_N2 rate depends on:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-NAT] [1 mark]

Benzene undergoes:

(Numerical Answer Type — Enter the numerical value)

Q64. [2-MCQ] [2 marks]

Hofmann elimination gives:

- (A) Most substituted alkene
 - (B) Least substituted
 - (C) Mixture
 - (D) No product
-

Q65. [2-MCQ] [2 marks]

Geometrical isomerism in:

- (A) 1-butene
- (B) 2-butene
- (C) 1-butyne
- (D) 2-butyne

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	D	1
2	1MCQ	A	1
3	1NAT	D	1
4	1MCQ	C	1
5	1NAT	B	1
6	1MCQ	B	1
7	1NAT	C	1
8	1MCQ	A	1
9	1NAT	B	1
10	1MCQ	C	1
11	1NAT	A	1
12	1MCQ	A	1
13	1NAT	C	1

14	1MCQ	C	1
15	1MCQ	B	1
16	1NAT	A	1
17	1NAT	B	1
18	1MCQ	B	1
19	1NAT	B	1
20	1NAT	B	1
21	1MCQ	A	1
22	1MCQ	A	1
23	1MCQ	A	1
24	1MCQ	B	1
25	1NAT	B	1
26	1MCQ	A	1
27	1MCQ	B	1
28	1MCQ	B	1

29	1MCQ	A	1
30	1MCQ	C	1
31	2NAT	A	2
32	2NAT	A	2
33	2MCQ	C	2
34	2NAT	B	2
35	2NAT	C	2
36	2MCQ	B	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2MSQ	A	2
41	2MSQ	A	2
42	2NAT	B	2
43	2NAT	B	2

44	2NAT	B	2
45	2NAT	A	2
46	2NAT	B	2
47	2MCQ	C	2
48	2NAT	B	2
49	2MCQ	D	2
50	2NAT	D	2
51	2MCQ	C	2
52	2MCQ	B	2
53	2MCQ	A	2
54	2MCQ	B	2
55	2NAT	B	2
56	2MCQ	A	2
57	2MCQ	C	2
58	2NAT	B	2

59	2NAT	A	2
60	2MCQ	B	2
61	2MCQ	C	2
62	2MCQ	B	2
63	2MCQ	D	2
64	2MCQ	B	2
65	2MSQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: D

Solution:

Perchloric acid (HClO_4) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q2.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q3.

Answer: D

Solution:

Very dilute: $[\text{H}^+]$ from water dominates. $[\text{H}^+]_{\text{total}} \approx 1.1 \times 10^{-7}$. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q4.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a) = f(b)$. Then $f'(c) = 0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q5.

Answer: B

Solution:

Hofmann (bulky base): gives least substituted (Hofmann) alkene.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q6.

Answer: B

Solution:

Grignard: reacts with CO_2 (carboxylic acid), carbonyls, epoxides. Reacts with protic solvents.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: easy

Q7.

Answer: C

Solution:

CN^- (strongest) $> \text{NH}_3 > \text{en} > \text{NO}_2^- > \text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$ (weakest).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q8.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q9.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q10.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q11.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q12.

Answer: A

Solution:

$$dG = -RT \ln(K) \Rightarrow K = \exp(-dG/RT) = \exp(5000/(8.314 \cdot 300)) = e^2.$$

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: difficult

Q13.

Answer: C

Solution:

$$y = A \cos(2x) + B \sin(2x). A=1, B=0. y(\pi/2) = \cos(\pi) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q14.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q15.

Answer: B

Solution:

$$2 = \exp(E_a/R \cdot (1/300 - 1/310)). E_a = R \ln(2)/(1/300 - 1/310) = 53.6 \text{ kJ/mol. Closest: 56.}$$

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q16.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q17.

Answer: B

Solution:

$$\Delta_n = 2 - 4 = -2.$$

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q18.

Answer: B

Solution:

Geometrical: restricted rotation around double bond. 2-butene has cis/trans.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q19.

Answer: B

Solution:

$$[H^+] = 10^{-2}. \text{pH} = -\log(10^{-2}) = 2.$$

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q20.

Answer: B

Solution:

SN2: bimolecular. Rate = $k[\text{substrate}][\text{nucleophile}]$. Both affect rate.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q21.

Answer: A

Solution:

Low-spin d_6 : t_{2g}^6 , all paired. Diamagnetic.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: difficult

Q22.

Answer: A

Solution:

$$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min.}$$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q23.

Answer: A

Solution:

Octahedral: t_{2g} (lower: d_{xy}, d_{yz}, d_{zx}) and e_g (higher: $d_{x^2-y^2}, d_{z^2}$).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: difficult

Q24.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: B

Solution:

Benzene: aromatic stability favors electrophilic aromatic substitution.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: easy

Q26.

Answer: A

Solution:

$$L\{t^s f(t)\} = -d/ds(F(s)). \quad F(s) = 1/(s+2). \quad -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: B

Solution:

$$t_{90} = t_{50} \cdot \log(10)/\log(2) = 10 \cdot 3.32 = 33.2 \text{ min.}$$

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q28.

Answer: B

Solution:

Werner: primary valency (oxidation state) + secondary valency (CN). Explained geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q29.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q30.

Answer: C

Solution:

$(P + a/V^2)(V - b) = RT$. Corrects for intermolecular forces (a) and finite volume (b).

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: difficult

Q31.

Answer: A

Solution:

$$\text{Area} = \pi r^2. \quad \text{Circumference} = 2\pi r.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \quad \text{Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: C

Solution:

6 NH₃ ligands: CN = 6. Octahedral geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q34.

Answer: B

Solution:

$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q35.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1, 0]^T$ is eigenvector for eigenvalue 1. $A^{50} [1, 0]^T = [1, 0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q40.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1 + 4 + 9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q41.

Answer: A

Solution:

Homogeneous ODE. Substituting $y = x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q45.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: B

Solution:

Werner: primary valency (oxidation state) + secondary valency (CN). Explained geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q47.

Answer: C

Solution:

CN^- (strongest) > NH_3 > en > NO_2^- > F^- > Cl^- > Br^- > I^- (weakest).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q48.

Answer: B

Solution:

$[\text{H}^+] = 10^{-2}$. $\text{pH} = -\log(10^{-2}) = 2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q49.

Answer: D

Solution:

Perchloric acid (HClO_4) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q50.

Answer: D

Solution:

Very dilute: $[\text{H}^+]$ from water dominates. $[\text{H}^+]_{\text{total}}$ approx 1.1×10^{-7} . pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q51.

Answer: C

Solution:

6 NH_3 ligands: $\text{CN} = 6$. Octahedral geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q52.

Answer: B

Solution:

$t_{90} = t_{50} * \log(10)/\log(2) = 10 * 3.32 = 33.2 \text{ min}$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: easy

Q53.

Answer: A

Solution:

$dG = -RT * \ln(K) \Rightarrow K = \exp(-dG/RT) = \exp(5000/(8.314 * 300)) = e^2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q54.

Answer: B

Solution:

Hofmann (bulky base): gives least substituted (Hofmann) alkene.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: easy

Q55.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: C

Solution:

CN^- (strongest) > NH_3 > en > NO_2^- > F^- > Cl^- > Br^- > I^- (weakest).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q58.

Answer: B

Solution:

$t_{90} = t_{50} * \log(10)/\log(2) = 10 * 3.32 = 33.2 \text{ min.}$

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: difficult

Q59.

Answer: A

Solution:

Octahedral: t_{2g} (lower: d_{xy}, d_{yz}, d_{zx}) and e_g (higher: $d_{x^2-y^2}, d_{z^2}$).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q60.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q61.

Answer: C

Solution:

$(P + a/V^2)(V - b) = RT$. Corrects for intermolecular forces (a) and finite volume (b).

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q62.

Answer: B

Solution:

Werner: primary valency (oxidation state) + secondary valency (CN). Explained geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q63.

Answer: D

Solution:

Very dilute: $[\text{H}^+]$ from water dominates. $[\text{H}^+]_{\text{total}}$ approx 1.1×10^{-7} . pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q64.

Answer: B

Solution:

Geometrical: restricted rotation around double bond. 2-butene has cis/trans.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q65.

Answer: B

Solution:

$\Delta n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 27 questions, 43 marks
Inorganic Chemistry: 11 questions, 18 marks
Chemistry: 10 questions, 15 marks
Physical Chemistry: 10 questions, 15 marks
Organic Chemistry: 7 questions, 9 marks

Difficulty Distribution:

Easy: 18 | Moderate: 37 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.